

Aadit Kannan

Robotics hardware, electromechanical systems, PCB design, CAD prototyping, manufacturing, and validation

Mechanical engineering and EECS student at UC Berkeley focused on compact robotic actuators, high-voltage electric vehicle hardware, lab instrumentation, manufacturing drawings, and controls-aware robot mechanisms.

Actuators

PRIMARY FOCUS

HV + PCB

SYSTEMS

CAD + DFM

METHODS

Bench + Fit-Up

VALIDATION

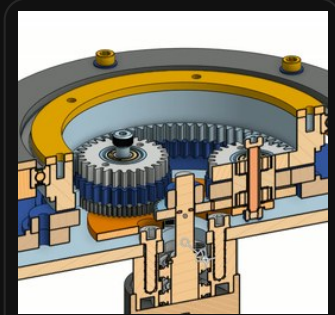
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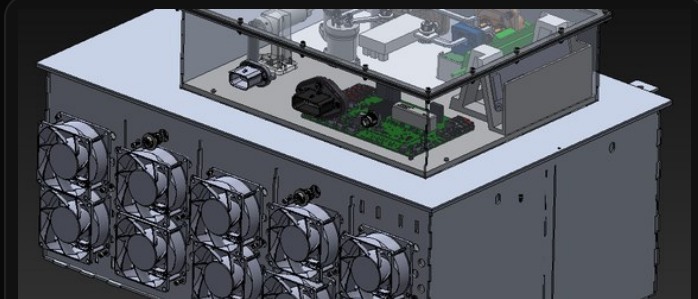
Portfolio generated from existing site pages and project assets, reframed for mechanical/hardware robotics roles.



Wolfram actuator prototype



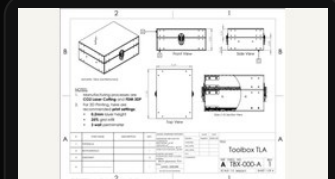
Compound planetary section



588 V accumulator CAD

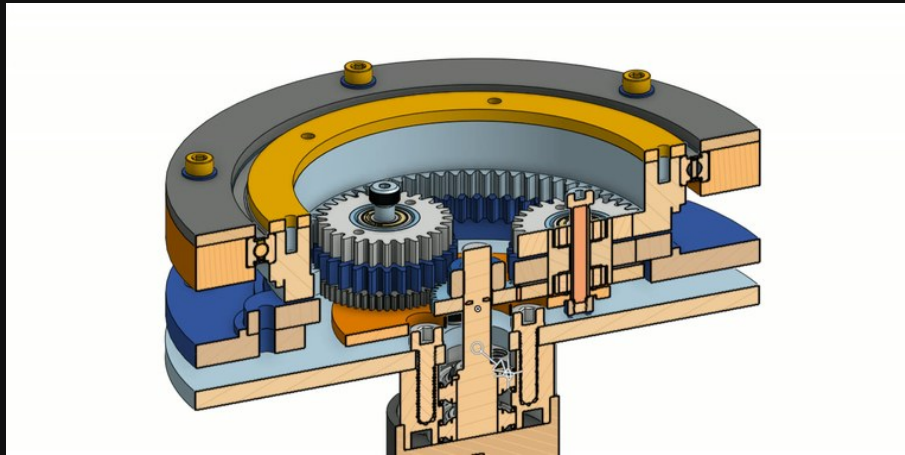


ns pulse schematic



GD&T drawing package

Compact Wolfrom Compound Planetary Actuator



Section view: driven sun, compound planets, fixed ring, output ring, bearings.

PROBLEM

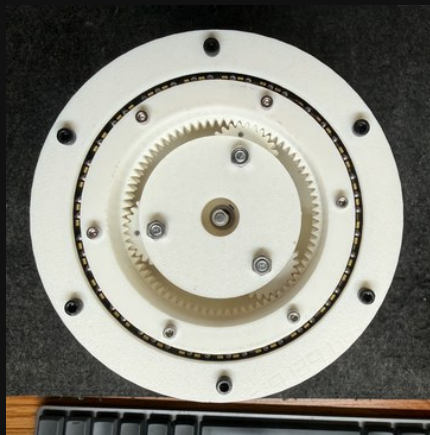
Humanoid-scale joints need compact, high-ratio torque transmission without stacking bulky conventional gearboxes. The design target was a compact actuator module that could be prototyped quickly while preserving a path toward stronger manufacturing.

MY ROLE

Owned the mechanical architecture and CAD: gearbox geometry, actuator stackup, bearing support, output ring splitting, bolt access, axial spacing, and prototype validation planning.

ENGINEERING WORK

- Designed Wolfrom-style compound planetary layout targeting approximately 50:1 reduction.
- Modeled sun gear, compound planets, fixed ring, output ring, carrier, shafts, bearings, and output hub.
- Iterated assembly sequence around fastener access, ring splitting, bearing stackups, and printability.
- Built printed prototypes to evaluate gear meshing, packaging, assembly order, and mechanical fit.



Printed prototype with internal ring and bearing support.



Axial package and ring/hub geometry.

Onshape

CAD

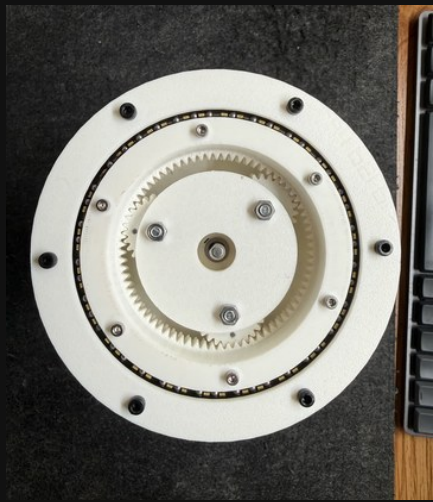
Actuator Design

Gearbox Packaging

Bearing Stackups

FDM/SLA Prototyping

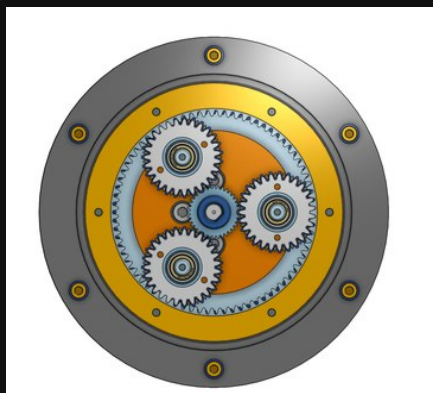
Wolfrom Actuator - Integration And Prototype Validation



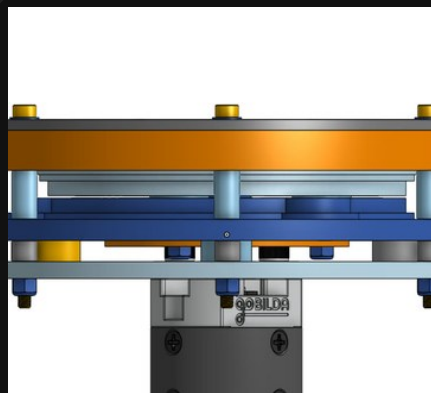
Internal stackup and ring gear access during assembly.



Bench prototype used for fit and gear-mesh evaluation.



Output-side packaging and fastener pattern.



Axial stack height and housing interfaces.

DESIGN CONSTRAINTS

- Gear clearance and alignment through the compound planet path.
- Output ring stiffness and bearing support without excessive axial growth.
- Assembly access for low-profile fasteners and ring/housing interfaces.
- Prototype manufacturability using printed parts before a metal/SLA hybrid version.

ARCHITECTURE

The motor drives the sun gear. One planet gear meshes with a fixed internal ring while the paired planet drives the output internal ring. That compound layout creates high reduction in one compact axial package, instead of stacking multiple gearbox stages.

STATUS

Current status: iterating printed prototypes and mechanical packaging. No production-ready or measured torque/efficiency claims; the near-term goal is clean prototype validation and a stronger manufacturable version.

~50:1

TARGET RATIO

humanoid

DOMAIN

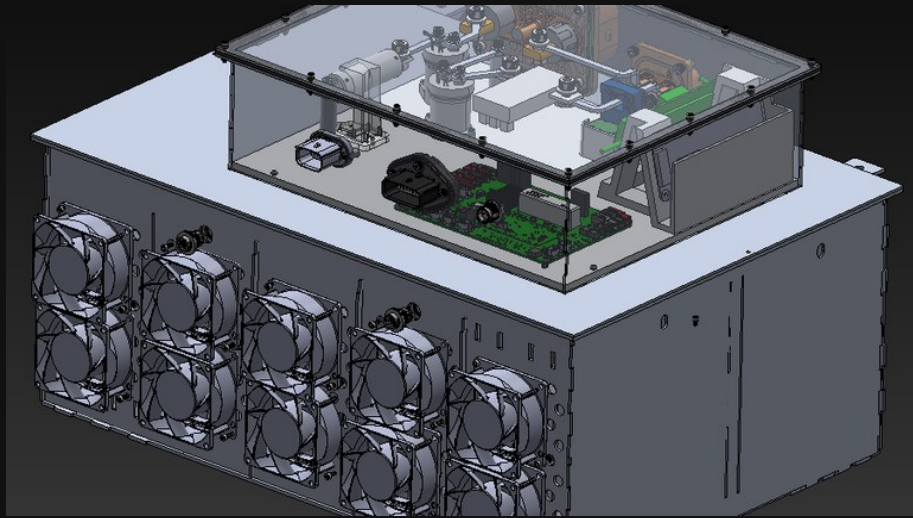
prototype

STAGE

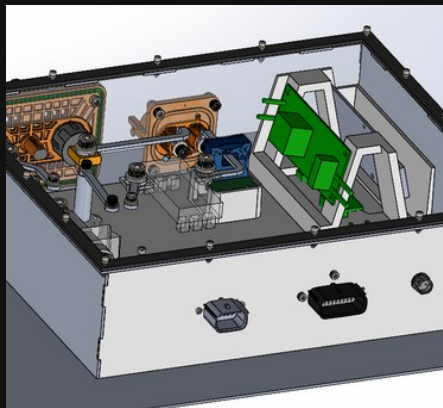
metal/SLA

NEXT

Formula Electric Berkeley - 588 V Accumulator And HV Electronics Attic



Accumulator CAD with HV electronics attic, service access, cooling, and enclosure packaging.



Electronics attic layout and enclosure-level packaging.



Hardware packaging view used for integration planning.

PROBLEM

A Formula SAE electric accumulator is a dense safety-critical electromechanical system: high-voltage energy storage, low-voltage electronics, mechanical enclosure constraints, airflow, serviceability, and rule compliance all collide in one package.

MY ROLE

Designing the custom high-voltage 588 V accumulator and electronics attic packaging for Formula Electric at Berkeley, with focus on integration between structural packaging, HV isolation, airflow, cable routing, and maintenance access.

ENGINEERING WORK

- CAD packaging for accumulator enclosure, electronics attic, boards, fans, service access, and fastener interfaces.
- Electromechanical layout decisions around isolation, connector access, cooling paths, and inspection/maintenance workflow.
- Mechanical integration of HV hardware with electronics constraints rather than treating the enclosure as a simple box.
- Design tradeoffs among stiffness, accessibility, manufacturability, and competition vehicle packaging.

SolidWorks

HV Packaging

Accumulator Design

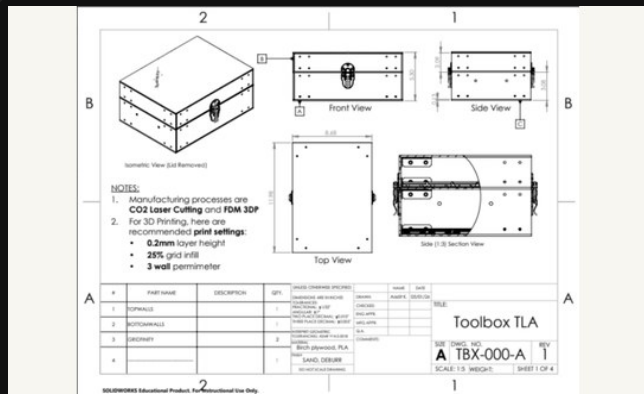
Electromechanical Integration

DFM

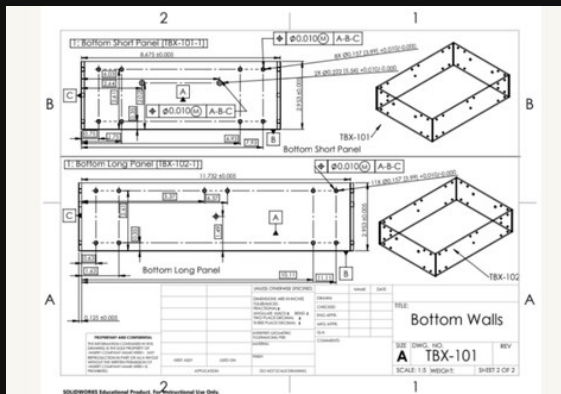
Serviceability

Formula SAE Electric

Custom Toolbox Design And Manufacturing



Assembly drawing defining lid, latch, hinge, handle, body, and insert interfaces.



Part drawing with datum references and positional tolerance callouts.

PROBLEM

This was a compact manufacturing exercise: take a physical assembly from CAD to production-style drawings, tolerancing, fabrication planning, inspection, and prototype assembly.

ENGINEERING WORK

- Modeled full assembly and generated part/assembly drawings.
- Applied ASME Y14.5-2018-style GD&T principles for datums and hole patterns.
- Designed around CO2 laser-cut plywood panels, FDM printed inserts, and fastened joints.
- Checked fit-up with calipers, fastener fit checks, squareness checks, and physical assembly validation.

RESULT

Prototype photos show the fabrication and fit-up validation stage. The value is the workflow: design intent, datums, functional tolerances, manufacturable drawings, and inspection feedback.



Prototype fit-up after fabrication with hinges, latches, handle, and fasteners.



Open prototype showing internal fit-up and hardware alignment.

SolidWorks

GD&T

ASME Y14.5

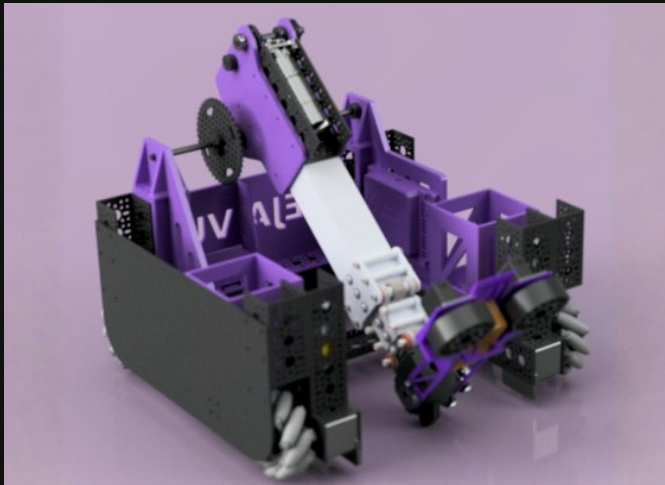
DFM

Laser Cutting

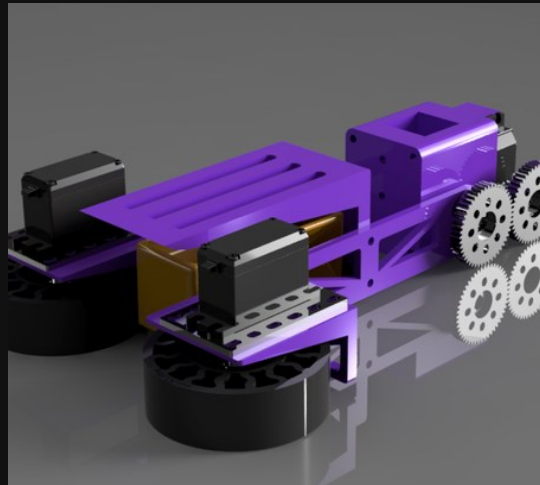
FDM

Inspection

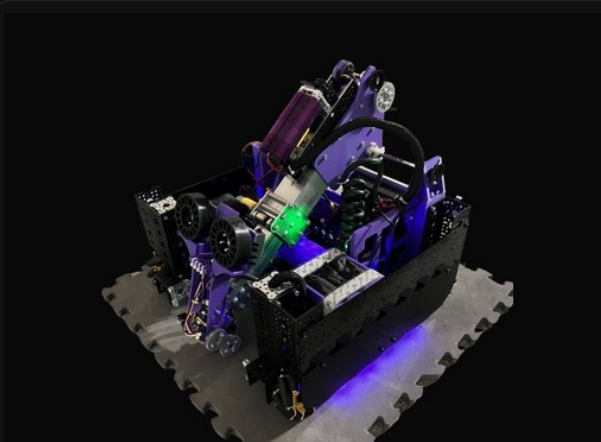
FIRST Robotics - Controls-Aware Mechanical Design



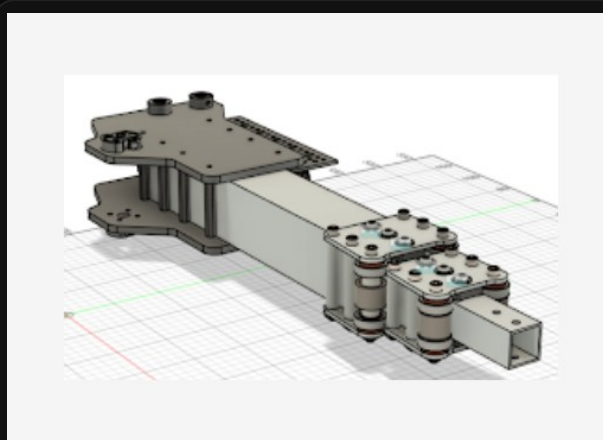
Deja Vu INTO THE DEEP robot CAD and subsystem packaging.



High-iteration intake mechanism for game-piece acquisition.



Competition robot hardware integration.



Linkage and end-effector packaging detail.

ROLE

Hardware Lead for FTC #13216, owning mechanical design, CAD management, manufacturing coordination, and driver-facing TeleOp code for competition robots.

ENGINEERING WORK

- Designed robot assemblies with hundreds of custom CAD parts across drivetrain, intake, transfer, lift, arm, end effector, and hang.
- Iterated intakes through multiple revisions based on game-piece behavior, reliability, and driver feel.
- Manufactured with printed structural parts, laser-cut plates, CNC-machined aluminum, and standard shop processes.
- Wrote Java TeleOp code for field-centric mecanum drive, mechanism sequencing, PID loops, and gamepad feedback.

SolidWorks

FTC SDK

Java

Mecanum

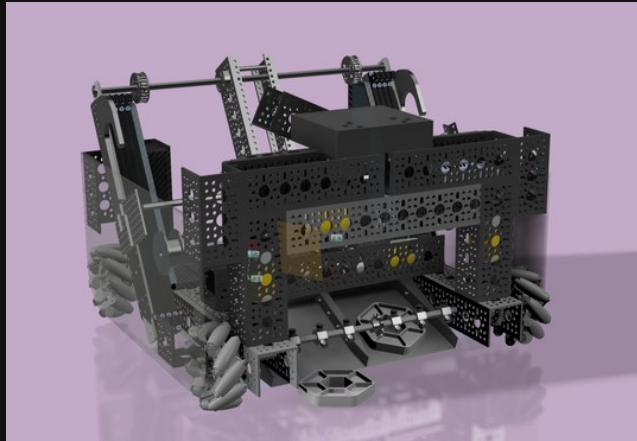
PID

Laser Cutting

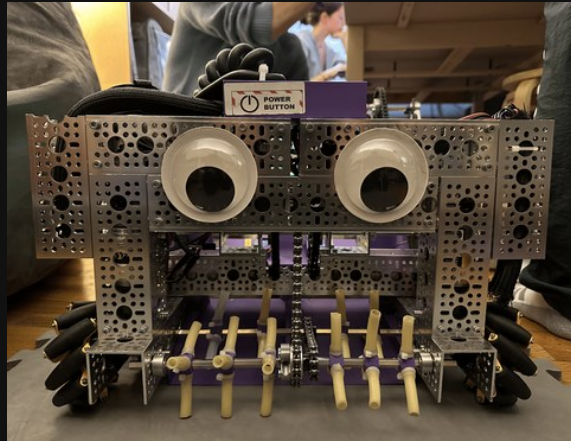
CNC

FEA

FIRST Robotics - Mechanism Iteration And Manufacturing



CENTERSTAGE robot CAD assembly with scoring mechanism layout.



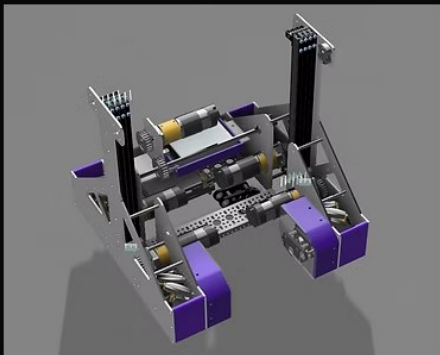
Physical robot build and mechanism packaging.

PROBLEM

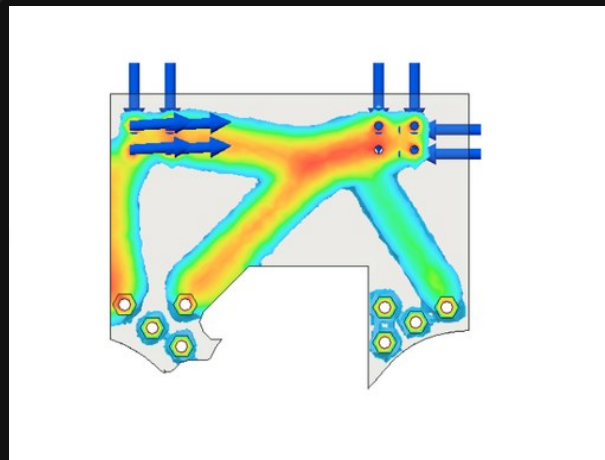
FTC robots compress mechanism design, manufacturing, controls, reliability, and rapid iteration into short build cycles. The useful skill transfer is not the game itself; it is design-build-test under constraints.

RESULT / STATUS

Built competition-ready robots with minimal mechanical failures during matches and advanced through qualifications into elimination rounds. These projects formed the base for later actuator, battery, PCB, and manufacturing work.



POWERPLAY-era robot hardware foundation.



Simulation/CAD workflow for drivetrain and mechanism planning.

WHAT TRANSFERS TO FIGURE

- Mechanisms designed around real drivers, real manufacturing limits, and repeated field failures.
- Controls-aware hardware decisions: geometry, compliance, sensing, sequencing, and serviceability.
- Comfort moving between CAD, fabrication, software, and match-side debugging.

Capability Map For Robotics Hardware Roles

Actuation + Mechanisms

- Compact gearbox architecture
- Bearing/output stackups
- Assembly sequence and prototype validation
- Controls-aware mechanism design

Electromechanical Systems

- 588 V accumulator packaging
- HV electronics attic layout
- Service access and cooling constraints
- Mechanical/electrical interface thinking

PCB + Lab Hardware

- KiCad schematic/layout workflow
- LTspice timing checks
- Oscilloscope bring-up planning
- Coax/BNC test interfaces

Manufacturing + Inspection

- Production-style drawings
- GD&T datum strategy
- DFM for available processes
- Calipers, fastener checks, fit-up validation

Links

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Selected optional systems

- PLDTracker: Python/Pandas dashboard for thin-film deposition metadata and characterization images.
- Leitmotif: real-time multimodal software system using vision, generative music, and persistent motif data.

Portfolio stance

The emphasis is practical robotics hardware: package the mechanism, design the interface, build the prototype, instrument the test, and learn from fit-up or bench data.